

# Reconstruction, not cosmetic smoothing

Learning outcomes: aggregate additive particle moments correctly; distinguish sampling error from genuine flow structure; explain prior-plus-observation reconstruction; preserve measured zero-frequency content; use a support warning without overstating its guarantees.

Suggested 90-minute class: 20 minutes of moment algebra, 15 minutes of the inverse problem, 30 minutes of the executable experiment, 15 minutes of physical audits and 10 minutes for assessment. Prerequisites are DSMC sampling, supervised splits, POD and the Week 10 moment hierarchy. Week 11 motivates why noisy derivatives can destroy feature identification.

Research source: Ehsan Roohi, Geometry-native machine learning reconstruction of DSMC moment fields with support monitoring, arXiv:2609.01637 (2026), <https://doi.org/10.48550/arXiv.2609.01637>. The author confirms submission to Journal of Computational Physics. This lecture does not call it a published JCP article.

The paper's cavity prior is a trained MambaIR restoration model; its cylinder estimator uses geometry-adapted, coupled heat-flux reconstruction. The CPU notebook is a transparent single-scalar spectral analog, not those trained models or a reproduction of their measured accuracy.

# From molecular velocities to the moment hierarchy

DSMC estimates a distribution in two spatial dimensions with three molecular-velocity components. Retained fields include number density, two bulk velocities, translational temperature, three pressure-tensor components and two heat-flux components. Heat flux is a central third-order moment, not simply a smoothed temperature gradient.

For simulator weights  $w$  and molecular velocity  $\mathbf{x}_i$ , accumulate  $C_0 = \sum(w)$ ,  $C_i = \sum(w \cdot x_{i_i})$ ,  $C_{ij} = \sum(w \cdot x_{i_i} \cdot x_{i_j})$ ,  $E_2 = \sum(w \cdot |\mathbf{x}_i|^2)$ , and  $F_i = \sum(w \cdot |\mathbf{x}_i|^2 \cdot x_{i_i})$ . Sum these additive quantities over the declared blocks first. Then compute  $u_i = C_i / C_0$  and  $J_i = F_i - u_i \cdot E_2 - 2 \cdot \sum_j (u_j \cdot C_{ji}) + 2 \cdot C_0 \cdot u_i \cdot |u|^2$ .

The physical heat flux requires the molecular-mass, cell-volume and sampling-time normalization. Separately centralising each block inserts different velocities into a nonlinear expression; averaging those central quantities is not generally equivalent to pooling raw accumulators first. The notebook demonstrates the discrepancy and verifies the pooled expression against direct centralisation of all particles.

Large raw terms may nearly cancel in  $J_i$ . A small bias in velocity or energy flux can therefore become a large relative heat-flux error. Visually smooth fields can still have the wrong amplitude or sign.

# Observations are noisy; references are too

Raw(B) denotes an estimate formed from B declared additive sampling blocks. For independent equal-variance blocks, averaging reduces variance by B. Correlation, unequal exposure, sampling resets and nonstationarity invalidate a naive block-count argument. Do not treat neighboring cells or overlapping cumulative outputs as independent replicates.

A high-budget independent reference is not the exact Boltzmann solution. Compare every method against the same reference and retain observation/reference independence. Raw(3) may be nested inside Raw(10); that should be disclosed, while neither should overlap its evaluation reference. Keep development priors, gain selection and support thresholds separate from held-out evaluation pairs.

For native cell areas A,  $\text{NRMSE} = \sqrt{\sum(A * (\text{estimate} - \text{reference})^2) / \sum(A * \text{reference}^2)}$ . Declare masks and denominators. A near-zero reference norm makes relative error unstable. Report seed-level results, not only a favorable average. The ratio of mean errors differs from the mean of individual ratios.

The notebook uses explicitly synthetic independent Gaussian blocks and an independent 80-block-equivalent noisy reference. Analytic truth remains available only to diagnose this manufactured experiment. These fields cannot establish the noise covariance or convergence rate of a real DSMC solver.

# A prior must listen to the current observation

Let  $T_g$  be a geometry-adapted linear transform,  $z_{\text{obs}} = T_g * m_{\text{obs}}$  and  $z_{\text{prior}}$  the prior coefficients. Reconstruct  $z_{\text{hat}} = z_{\text{prior}} + H_g * (z_{\text{obs}} - z_{\text{prior}})$ . The operator's singular values are bounded between zero and one so it does not amplify the coefficient residual. Zero transfer returns the prior; identity transfer returns the observation before additional constraints.

For a scalar cavity DCT, a Wiener-type gain is  $G = \text{signal\_power} / (\text{signal\_power} + \text{noise\_power}/B)$ . Set the zero-frequency gain to one. A trusted historical shape can suppress sampling noise, but the current observation anchors changes in amplitude and mean. Replacing that observation with the prior can erase real departures from development data.

The paper uses a learned MambaIR prior for the cavity and a separately fitted cylinder transfer. Our notebook estimates a mean prior and scalar spectral gains from development draws. That is an interpretable training exercise, not a MambaIR implementation. Gains and Gaussian-filter widths are frozen before held-out evaluation.

For equal-area cells, restore the observation's arithmetic mean after reconstruction. For native unequal-area cells, restore its area-weighted Cartesian mean. This constraint preserves a measurement, not an exact truth; it does not guarantee positivity, full moment consistency or conservation by itself.

## Geometry and identifiable information

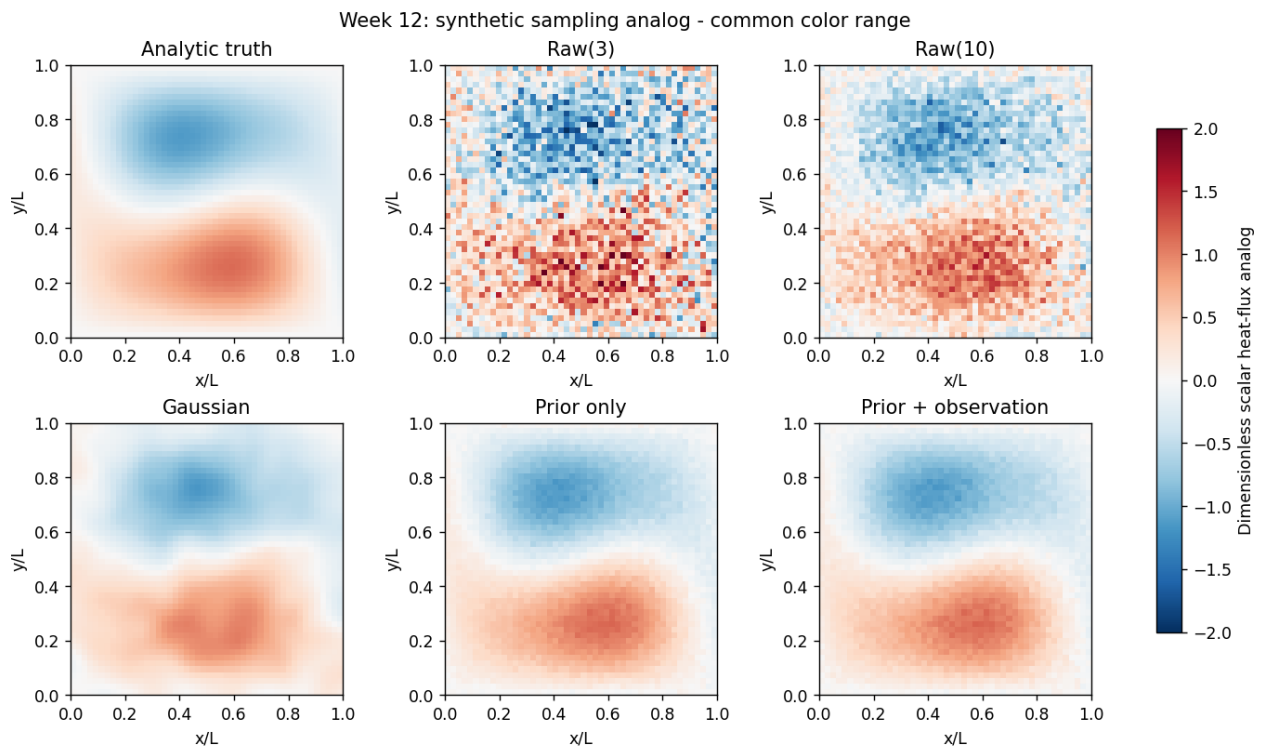
Around a cylinder, define the outward normal  $n=(\cos(\theta),\sin(\theta))$  and tangent  $t=(-\sin(\theta),\cos(\theta))$ . Rotate  $q_n=q_x\cos(\theta)+q_y\sin(\theta)$  and  $q_t=-q_x\sin(\theta)+q_y\cos(\theta)$ . The inverse is  $q_x=q_n\cos(\theta)-q_t\sin(\theta)$ ,  $q_y=q_n\sin(\theta)+q_t\cos(\theta)$ . Test the round trip numerically before filtering.

The paper transforms fluid-only normal/tangential fields in cylinder-centred coordinates, learns bounded two-component spectral transfer, returns to native Cartesian fields and restores the measured area-weighted means. Do not interpolate through the solid or confuse a smooth visualization grid with solver resolution. The Cartesian classroom grid does not implement this full cylinder treatment.

Energy conservation constrains divergence of heat flux, not both components uniquely. For any smooth  $\psi$ ,  $\delta q=(d\psi/dy,-d\psi/dx)$  has zero divergence. Adding it changes heat flux without changing its energy-residual contribution. The notebook verifies this derivative identity.

Thus a small energy residual cannot uniquely identify heat flux. Use current observations, independent references, geometry and physical diagnostics together. Do not enforce Fourier's law as ground truth in strongly non-equilibrium rarefied flow.

# Compare fields and failure modes



This executed figure uses one synthetic scalar heat-flux analog, a common color range and no research DSMC data. Methods are Raw(3), Raw(10), a validation-selected Gaussian filter, prior-only and prior plus observation. The notebook reports values outside the display range so saturation cannot silently hide noise extremes.

Judge reference NRMSE, analytic-truth error, gradient error and mean change. For real shocks add front position, thickness and peak bias; for a cylinder add near-wall normal heat flux and integral surface quantities. A global error can hide a serious local failure. No method is declared superior before the held-out results are read.

The notebook's analytic truth is a diagnostic convenience. Research tests instead need credible independent or leave-one-seed-out references, matched sampling budgets and explicit uncertainty limits. Noise reduction alone does not justify a computational speedup claim.

# Support monitoring and abstention

A new field may lie outside the range in which the prior and residual transfer were qualified. The paper monitors nine components in three geometry zones using observation residual power relative to development noise. Its gain-envelope score is calibrated on development units; the stated rule is a heuristic, not a universal finite-sample coverage guarantee.

The notebook deliberately implements a simpler one-field residual/noise score. The maximum validation score is frozen before a shifted observation is examined. No test reference or test error enters the decision. Label this an illustrative heuristic, not the paper's 27-component-zone monitor or a calibrated probability of failure.

If the score exceeds the threshold, abstain from presenting the reconstructed field as supported. Keep the raw estimate, collect additional independent samples, or build new in-condition development evidence. Do not silently retune the prior after looking at the test reference. A within-envelope score alone does not prove accuracy.

Discussion: a mean-preserving estimator can carry a noisy or out-of-condition mean unchanged. What distinct roles do data consistency, support monitoring and independent physical validation play? Why is an attractive contour insufficient evidence for any of them?

# Assessment and research handoff

Submit the pooled-versus-separate central-moment check, the frozen experiment protocol, all six synthetic test-case metrics, a contour comparison, and the support decision for a shifted observation. Include a case where the simplest baseline is competitive or better. Explain which information the prior supplies and which information the observation preserves.

Extensions: introduce temporal correlation and measure the failure of the  $1/B$  noise law; add a narrow layer and quantify smoothing bias; implement normal/tangential rotation and area-weighted mean restoration; design independent references for a real DSMC run. These are extensions, not already-executed article reproductions.

Run notebooks/week12/W12\_DSMC\_Moment\_Reconstruction.ipynb using CPU dependencies in FlowMLLab or its Colab launcher. A normal notebook run leaves retained evidence unchanged. References to the paper concern its formulation; every metric generated by this notebook concerns only the labelled synthetic analog.

Reading: Ehsan Roohi, Geometry-native machine learning reconstruction of DSMC moment fields with support monitoring, arXiv:2609.01637, <https://doi.org/10.48550/arXiv.2609.01637>. Lecture wording and classroom code are original, AI-assisted FlowMLLab additions grounded in the author's manuscript. No original research data, solver or checkpoint is fabricated or silently substituted.